

Introduction to Robotics

T.U.B.U.R.A.N. Learning Module

What is a Robot?

A robot is a machine that can sense its environment, process information, and take physical action based on that information. Every robot, no matter how simple or complex, is built around three core building blocks.

1. Sensors (Input)

Sensors allow a robot to perceive its surroundings. Common examples include light sensors, distance sensors (ultrasonic or infrared), touch sensors, and cameras. Sensors convert real-world signals into data the robot's controller can understand.

2. Controllers (Processing)

The controller is the 'brain' of the robot. It receives data from sensors, processes it according to a program, and decides what action the robot should take. Common controllers used in student robotics kits include Arduino and micro:bit boards.

3. Actuators (Output)

Actuators are the parts that let a robot act on its environment: motors that spin wheels, servo motors that move arms, or LEDs that light up. Actuators translate the controller's decisions into physical movement or output.

Putting It Together

A simple example: a light sensor detects darkness, the controller decides to turn on an LED, and the LED (the actuator) lights up. This sense-think-act cycle is the foundation of all robotics, from simple line-following robots to complex industrial arms.